

Ishan Sehgal

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Profile

Aspiring robotics engineer with a focus on autonomous systems and navigation. Recent Electronics and Communication graduate from Guru Nanak Dev University, Amritsar, with hands-on experience in developing real-world robotic solutions, exploring novel approaches to localization, and implementing reliable control systems.

Skills

Programming Languages: C++, Python; **Robotics Frameworks:** ROS, ROS 2, Open-RMF, MoveIt2; **Hardware:** Raspberry Pi, YD LiDAR, ESP32, Depth cameras, GPS, 3D-Lidars, Robotics Manipulators, Motor Encoders; **Software Tools:** Fusion 360, Gazebo, SLAM, Nav/Nav2, Behavior Trees, Linux, Stereo Odometry, Sensor Fusion

Experience

Associate Robotics Engineer at Rigbetel Labs LLP

June 2025

- Developed a social robot with autonomous navigation capabilities, implementing custom Nav2 plugins and behavior trees.
- Deployed SLAM and AMCL (Beluga) for robust localization and explored sensor fusion approaches using stereo depth odometry, RTAB-Map, and EKF.
- Worked with Unitree G1 humanoid robot, contributing to advanced robotics research and development.

Internship at Rigbetel Labs LLP

October 2024 - June 2025

- Conducted hands-on projects with real robots, gaining practical experience in robotics implementation and deployment.
- Successfully converted two AGVs (Automated Guided Vehicles) to fully autonomous AMRs (Autonomous Mobile Robots).
- Executed multiple robotics projects involving real-world applications and autonomous navigation systems.

EYSIP 2024 Summer Internship at e-Yantra

May 2024 - July 2024

- Developed behavior trees for autonomous eBot and UR5 arm, improving reliability by reducing manual resets and downtime.
- Designed fail-safe mechanisms and implemented auto-localization, enabling Nav2 stack initialization from any location.

E-YRC 2023 at IIT Bombay

September 2023 - January 2024

- Advanced to Stage 2, securing AIR-13 in the Cosmo Logistics theme.
- Developed an autonomous sorting and packaging system using a robotic arm and mobile robot, optimizing algorithms.

Projects

Autonomous eBot Project

- Developed an autonomous eBot using ROS 2 Foxy, integrating Ylidar, motor encoders, and an ASUS i5 motherboard.
- Implemented SLAM for navigation and obstacle avoidance, achieving consistent localization accuracy across dynamic environments.
- GitHub link: E-bot

TARS Robot Project

- Built a working version of TARS robot inspired by Interstellar, implementing ROS 2 architecture for control and coordination.
- GitHub link: TARS-ROS2

Awards & Achievements

- Final stage qualifier of EYRC-2023 (top 15 teams).
- Secured first position in the 'I am Engineer' contest at university.
- Secured first prize in Science Exhibition TechNit' 16.
- Member of Technical Club at Guru Nanak Dev University under ESF.

Education

Guru Nanak Dev University Amritsar

2021-2025

- B.Tech Electronics and Communication
- CGPA: 7.23

Secondary Education

- MGN Public School Jalandhar
- 12th Percentage: 85.8%

2018-2021